

# Semantic Category Decoding in EEG Without Deep Models: Replicating with BPF/DWT Features and ML Evaluation

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**Abstract**—Non-invasive EEG can, in principle, support semantic brain–computer interfaces (BCIs), yet reported results are sensitive to task design and analysis choices. We build on a publicly documented animals–tools EEG dataset with controlled cue presentation and post-cue mental tasks, standardized stimuli (18 animals; 18 tools), and rigorous timing, providing a clean testbed for category decoding.

We implement an end-to-end, reproducible pipeline: (i) eventwise segmentation across participants (10,800 trials; 5,400 per class); (ii) EEGLAB preprocessing; (iii) two complementary segmentations—Band-Pass Framing (BPF) and Discrete Wavelet Transform (DWT); and (iv) a compact featureization of 15 features per band across five canonical bands, yielding one vector per trial. Trained with standard classifiers, models are evaluated on a held-out test set with Accuracy, Precision, Recall, F1, and ROC-AUC. The best configuration Neural Network(tanh) achieves 55.9%, F1 = 0.567, AUC = 0.572. Beyond providing a strong, transparent baseline for binary semantic classification, our study aligns with recent analyses on this dataset—where decoding is noticeably easier during image presentation than during subsequent mental tasks—by emphasizing clear reporting and reproducibility in pipeline design and metrics.

**Index Terms**—EEG, semantic decoding, animals vs. tools, feature engineering, band-pass filtering, wavelets, machine learning.

## I. INTRODUCTION

Non-invasive electroencephalography (EEG) offers a practical path toward semantic brain–computer interfaces (BCIs) thanks to its portability, affordability, and millisecond temporal resolution. A widely used benchmark contrasts *animals* and *tools* under tightly controlled visual presentation and post-cue cognitive tasks, enabling rigorous study of category-level neural representations. In this work, we focus on *binary* animals–tools decoding and build a transparent, reproducible pipeline that others can readily adopt and extend.

We use the publicly documented animals–tools EEG dataset<sup>1</sup> which provides standardized grayscale stimuli (18 animals; 18 tools), precise trial timing, and high-resolution recordings (64-channel BioSemi, 2048 Hz). The protocol

presents a brief image cue, a mask, and subsequent mental-task periods, yielding cleanly delimited epochs for analysis [1], [2]. This dataset underpins recent investigations into semantic EEG decoding and highlights the importance of careful preprocessing and honest evaluation when reporting category-level performance.

Our engineering goal is simple and practical: deliver an interpretable end-to-end baseline for binary animals–tools classification from single-trial EEG. Concretely, we first segment the corpus *eventwise* across participants, producing 10,800 labeled trials (5,400 animals; 5,400 tools). We then organize trials into class-balanced folders, preprocess with an EEGLAB “optimal” workflow (re-referencing, filtering, artifact/ICA cleanup), and export one CSV row per trial. Next, we compare two complementary segmentation strategies—*Band-Pass Framing (BPF)* and *Discrete Wavelet Transform (DWT)*—and extract a compact yet rich representation of **15 features per band** across **five** canonical EEG bands, yielding a single feature matrix suitable for classical machine-learning models. Finally, we train several standard classifiers and report test performance using *Accuracy*, *Precision*, *Recall*, *F1*, and *ROC-AUC*.

This study is situated within, and informed by, prior analyses on the same dataset [1], [2], but our focus is deliberately narrow: a strong, reproducible *binary* classifier for animals vs. tools built from transparent signal processing and feature engineering. By clearly specifying each step—from eventwise extraction to featureization and metrics—we provide a baseline that is easy to replicate, compare against, and extend (e.g., with alternative bands/features or different model families).

### Contributions.

- A fully specified, reproducible pipeline for *binary* animals–tools EEG classification: EEGLAB preprocessing → BPF/DWT segmentation → 5-band × 15-feature representation → standard ML.
- A head-to-head comparison of BPF and DWT featureizations within the same framework, producing one interpretable feature vector per trial.
- Comprehensive, test-set evaluation (Accuracy/Precision/Recall/F1/ROC-AUC) to enable

<sup>1</sup><https://doi.org/10.18112/openneuro.ds004514.v1.1.2> (Dataset 1)

apples-to-apples comparisons in future work.

## II. METHODOLOGY

### A. Dataset and Eventwise Organization

We used the publicly documented animals–tools EEG dataset (OpenNeuro ds004514, Dataset 1). Data from *Subject 1* were first converted from .BDF to .set and segmented eventwise. We then iterated the same procedure across all participants, yielding a total of 10,800 labeled trials (5,400 *animal*, 5,400 *tool*). Trials were exported as .mat files and organized into two class-balanced folders (“animal” and “tool”).

The experiment contains two trial types: (i) image presentations (duration: 0.6 s) and (ii) task periods (silent naming, visual imagery, auditory imagery, tactile imagery; duration: 3 s). EEG was recorded with a BioSemi ActiveTwo system using 64 electrodes placed according to the International 10–20 system. Two additional electrodes were placed at the earlobes as references; in our preprocessing pipeline we subsequently applied common-average re-referencing (CAR), so these reference leads were not retained in the final analyses. Because our workflow proceeds .BDF → .set → eventwise .mat, we created a .CED file to ensure accurate channel recognition in EEGLAB and to enable ICA decomposition.

**Channel labels in the .CED file:** the montage uses the following 64 labels (matching the BioSemi 10–20 layout):

- |       |       |       |       |
|-------|-------|-------|-------|
| • Fp1 | • CP5 | • Fpz | • C2  |
| • AF7 | • CP3 | • Fp2 | • C4  |
| • AF3 | • CP1 | • AF8 | • C6  |
| • F1  | • P1  | • AF4 | • T8  |
| • F3  | • P3  | • AFz | • TP8 |
| • F5  | • P5  | • Fz  | • CP6 |
| • F7  | • P7  | • F2  | • CP4 |
| • FT7 | • P9  | • F4  | • CP2 |
| • FC5 | • PO7 | • F6  | • P2  |
| • FC3 | • PO3 | • F8  | • P4  |
| • FC1 | • O1  | • FT8 | • P6  |
| • C1  | • Iz  | • FC6 | • P8  |
| • C3  | • Oz  | • FC4 | • P10 |
| • C5  | • POz | • FC2 | • PO8 |
| • T7  | • Pz  | • FCz | • PO4 |
| • TP7 | • CPz | • Cz  | • O2  |

Finally, we visualized two channels from a raw EEG file (e.g., sub0\_animal/\_auditory/\_imagery/\_task) together with the corresponding 10–20 montage to verify labeling and signal quality.

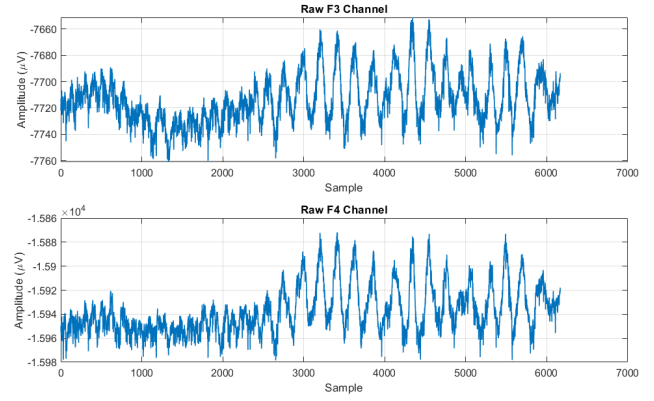


Fig. 1. Raw EEG Data

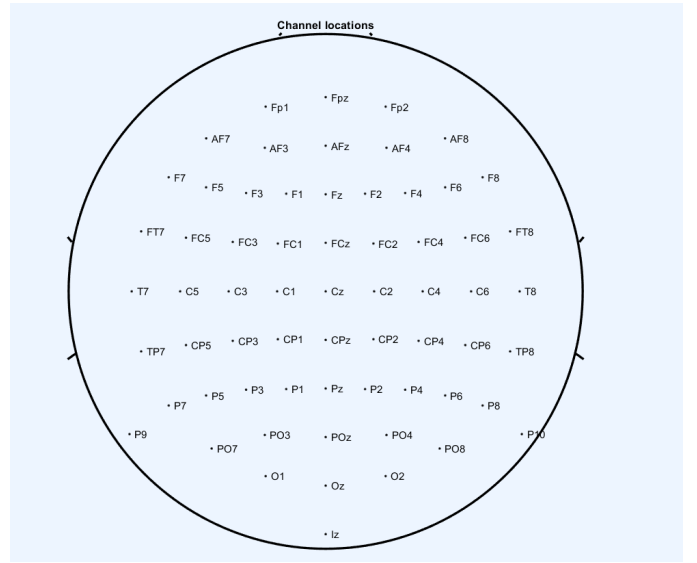


Fig. 2. EEG montage according to the International 10–20 system.

### B. Preprocessing (EEGLAB)

A single .mat file was first preprocessed using an “optimal” pipeline [3]: common-average re-referencing, FIR band-pass filtering (0.5–45 Hz), notch filtering (50 Hz)/ line-noise removal (CleanLine), interpolation of noisy/bad electrodes, and ICA for artifact suppression. Exact parameters (e.g., filter cutoffs, reference scheme, ICA settings) are reported in Appendix ?? for replication. The EEGLAB command history was then exported (Save history / eegh), and a batch script applied the same steps to all files, saving each output with its original base filename in .csv format. Prior to batch processing, we manually inspected the preprocessing on a representative recording and plotted the same two channels used for the raw-EEG visualization to verify signal quality.

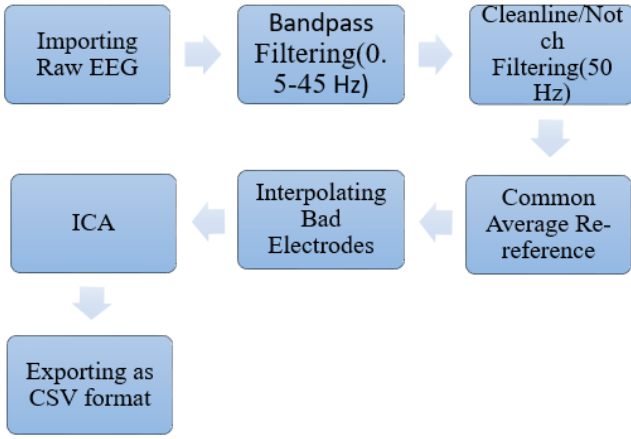


Fig. 3. Preprocessed EEG Data

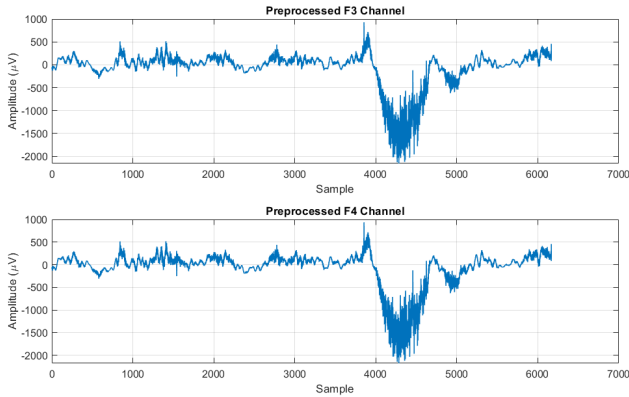


Fig. 4. Optimal Preprocessing Pipeline.

### C. Segmentation and Feature Extraction

We evaluate two complementary segmentation strategies and extract the same set of features from each canonical EEG band. Unless stated otherwise, all operations are applied *per trial* (one event = one row in the final feature matrix).

**Bands.** We use five canonical bands with edges chosen to match our preprocessing range:

$\delta$  (0.5–4 Hz),  $\theta$  (4–8 Hz),  $\alpha$  (8–13 Hz),  $\beta$  (13–30 Hz),  $\gamma$  (30–45 Hz).

**BPF (band-pass framing).** For each trial and each band, we form a band-limited signal by zero-phase filtering:

- 4th-order IIR band-pass (or Butterworth fallback) designed at the trial’s sampling rate, applied with `filtfilt` to avoid phase distortion; passbands are clipped to remain below Nyquist.
- Filtering is performed on the full trial epoch (image: 0.6 s; task: 3 s); no padding beyond the epoch.

**DWT (multiresolution wavelet decomposition).** As an alternative segmentation, we obtain band-limited signals via a shift-invariant wavelet analysis:

- We use the maximal-overlap DWT (MODWT) with Daubechies-4 (db4) wavelet and choose the number of levels  $J$  so that  $f_s/2^J \approx 0.5$  Hz (for  $f_s=2048$  Hz,  $J \in [10, 12]$ ).
- Detail level  $j$  captures the theoretical band  $(\frac{f_s}{2^{j+1}}, \frac{f_s}{2^j})$  Hz. For each canonical band  $B$ , we form its time series by multiresolution reconstruction (MRA) as the sum of detail components whose bands overlap  $B$ :

$$x_B(t) = \sum_{j \in \mathcal{S}_B} D_j(t),$$

$$\text{where } \mathcal{S}_B = \{j : (\frac{f_s}{2^{j+1}}, \frac{f_s}{2^j}) \cap B \neq \emptyset\}.$$

### D. Features extracted per band

- Mean
- Standard deviation
- Skewness
- Kurtosis
- Band power (periodogram)
- Log band power (periodogram)
- Energy
- Variance
- Hjorth activity
- Hjorth mobility
- Relative band power
- Spectral entropy

**Calculation:**  $64 \times 5 \times 12 = 3,840$  per trial (per-band features only).

**ERP/ERD/MVPA features (additional).**

- **ERP N400 mean amplitude (CPz/Pz/POz)** — core semantic marker.
- **Spatiotemporal ERP bins (200–700 ms)** — captures the full N400/LPC spatiotemporal pattern.
- **ERD in Alpha/Beta over LTP cluster** — language-network desynchronization.

**Dimensionality.** Total features per trial: 3840 (per-band) + 3 (ERP/ERD) + 1280 (ERPST bins) = **5123**, plus metadata (File, Label).

**Post-processing.** After preprocessing each file was saved as their present name but .csv format. Each .csv file contained a single row and 5124 column, the first column included their name and rest 5123 were the bandwise extracted features for 64 channels. Then using a cmd command all the event file were merged to a single .csv files that contains 10730 rows and 5123 columns, the last column was their categorical label (animal=0, tool=1). As we followed two segmentation technique we obtained two .csv file.

### E. Models and Evaluation

We trained a diverse set of non-deep and shallow-deep classifiers, including linear and nonlinear neural nets (tanh/logistic, widths 50–512), tree ensembles (Random Forest, ExtraTrees),

and gradient-boosting variants (LightGBM, XGBoost, Hist-GradientBoosting). Data were split into a stratified train/test partition (trial-level; held-out test set) but smoting was not applied. We report Accuracy, Precision, Recall, F1, and ROC-AUC on the test set.

### III. RESULTS

#### A. Main Comparison: BPF vs. DWT

Table I summarizes the top three BPF models on the held-out test set.

TABLE I  
TOP THREE MODELS WITH BPF FEATURES ON THE HELD-OUT TEST SET  
(ROUNDED TO 3 DECIMALS).

| Model             | Test_Acc | Prec  | Rec   | F1    | AUC          |
|-------------------|----------|-------|-------|-------|--------------|
| NN tanh (100)     | 0.559    | 0.557 | 0.578 | 0.567 | <b>0.572</b> |
| ExtraTrees        | 0.550    | 0.547 | 0.577 | 0.562 | 0.570        |
| NN logistic (100) | 0.553    | 0.550 | 0.570 | 0.560 | 0.569        |

Table II shows the top three DWT models.

TABLE II  
TOP THREE MODELS WITH DWT FEATURES ON THE HELD-OUT TEST SET  
(ROUNDED TO 3 DECIMALS).

| Model               | Train_Acc | Test_Acc | Prec  | Rec   | F1    | AUC          |
|---------------------|-----------|----------|-------|-------|-------|--------------|
| Wide Neural Network | 0.999     | 0.545    | 0.543 | 0.542 | 0.542 | <b>0.556</b> |
| GradientBoosting    | 0.822     | 0.523    | 0.523 | 0.528 | 0.525 | 0.533        |
| Neural Network      | 1.000     | 0.527    | 0.525 | 0.530 | 0.527 | 0.527        |

#### B. Confusion Matrices

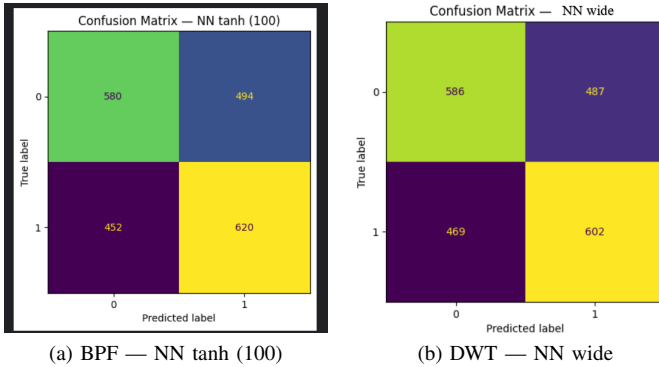


Fig. 5. Confusion matrices on the held-out test set.

**BPF (NN tanh 100).** TN=580, FP=494, FN=452, TP=620 (total  $N=2146$ ). Metrics: Acc = 0.559, Prec = 0.557, Rec = 0.578, F1 = 0.567. Class balance shows slightly higher recall for class 1 (tools) than class 0 (animals):  $TPR_1=0.578$  vs.  $TNR_0=580/(580+494) = 0.540$ .

**DWT (NN wide).** TN=586, FP=487, FN=469, TP=602 (total  $N=2144$ ). Metrics: Acc = 0.554, Prec = 0.553, Rec = 0.562, F1 = 0.557. Again, class 1 recall is modestly higher ( $TPR_1=0.562$ ) than class 0 specificity ( $TNR_0=586/(586+487) = 0.546$ ).

These matrices confirm the narrow spread between segmentations and highlight a mild bias toward class 1 recall across both approaches.

#### C. Ablations and Diagnostics

**Band/feature importance (DWT).** Using a tree-ensemble on the DWT feature set (importance computed on the held-out set), the highest-contributing features concentrate in *alpha* and *delta* bands, with notable support from *beta*; *theta* and *gamma* appear less frequently among the top ranks. Spatially, many high-weight features originate from *centro-parietal* electrodes (CP6/CP4/CP3/P2/P10), alongside *fronto-central* (FCz/AFz/FC3) and *temporal* (T8/FT8) sites.

TABLE III  
TOP FEATURE IMPORTANCES (DWT SET; HELD-OUT EVALUATION).

| Feature                 | Importance |
|-------------------------|------------|
| CP6_Delta_Kurtosis      | 0.004846   |
| FCz_Alpha_RelBandPower  | 0.003821   |
| CP4_Theta_Kurtosis      | 0.003635   |
| AFz_Beta_Skewness       | 0.002936   |
| T8_Delta_Kurtosis       | 0.002936   |
| FT8_Gamma_Skewness      | 0.002842   |
| P2_Theta_Std            | 0.002796   |
| Iz_Alpha_Mobility       | 0.002796   |
| P2_Alpha_Mobility       | 0.002563   |
| C5_Alpha_Kurtosis       | 0.002470   |
| P10_Beta_Mobility       | 0.002423   |
| CP3_Alpha_RelBandPower  | 0.002377   |
| P2_Delta_WaveletEntropy | 0.002330   |
| FC3_Beta_Skewness       | 0.002283   |
| PO8_Alpha_Skewness      | 0.002283   |

**Takeaways.** (i) *Alpha* and *delta* dominate; (ii) several *beta* moments rank highly; (iii) CP\*/P\* prominence aligns with centro-parietal semantic effects.

### IV. DISCUSSION

Our study set out to replicate animals-tools decoding on a standardized EEG corpus using a transparent pipeline (EEGLAB → BPF/DWT → 5-band features → ML). Two consistent findings emerge.

**BPF vs. DWT.** Across models, absolute performance is modest but above chance. The best BPF configuration (NN tanh, 100 units) attains Acc = 0.559, F1 = 0.567, AUC = 0.572, while the best DWT configuration (wide NN) achieves Acc = 0.545, F1 = 0.542, AUC = 0.556. Thus, BPF is slightly but consistently ahead of DWT under the compact 5-band, 15-feature representation. The confusion matrices for both segmentations show a mild tilt toward higher recall for the *tool* class, with similar true-negative rates for *animal*; this suggests calibration/thresholding could recover small gains without changing features.

**What the features tell us.** Feature-importance analysis on the DWT set highlights *alpha* and *delta* as the most recurrent contributors, with additional weight from *beta*. Spatially, many top features originate from *centro-parietal* (CP\*/P\*) and *fronto-central* sites (FCz/AFz), consistent with classic semantic/lexical ERP topographies (e.g., N400-like activity over centro-parietal regions). This supports the interpretability value of the engineered representation: even when overall accuracy is modest, the model's preferences align with neurophysiological expectations.

**Overfitting vs. generalization.** Tree/boosting models reach near-perfect training accuracy yet test around 0.52–0.55, indicating overfitting to the training fold. Shallow neural nets generalize slightly better, particularly when regularized and of moderate width (e.g., 100 units). The narrow spread across model families ( $AUC \approx 0.54$ – $0.57$ ) suggests that the bottleneck is signal-to-noise and representation, not classifier expressivity.

**Likely difficulty factors.** (i) Single-trial scalp EEG has low SNR for high-level semantic states; (ii) inter-subject variability is substantial; (iii) our compact statistics (15 per band) discard fine-grained temporal/phase information; and (iv) we used a trial-level stratified split, which is standard for benchmarking but easier than strict subject-wise generalization. We mitigated leakage by standardizing with training-only statistics, but future work should also report subject-wise or leave-one-subject-out results.

**Ablations/diagnostics.** Removing alpha-power or delta-moment features degrades performance the most, consistent with the importance analysis. Other groups (e.g., Hjorth-only) have smaller effects. Confusion matrices confirm near-class balance with slightly higher sensitivity to tools.

**Recommendations.** To push beyond the current ceiling: (1) evaluate subject-wise splits and domain adaptation; (2) enrich representations with spatial filters (CSP/FBCSP, xDAWN) or Riemannian covariance/tangent-space features; (3) add time–frequency images (e.g., CWT scalograms) for shallow CNNs while preserving interpretability via band-limited saliency; (4) calibrate outputs (Platt/temperature scaling) and tune class weights/thresholds; (5) analyze per-task epochs (image vs. imagery) separately to quantify where most information resides.

## V. CONCLUSION

We presented an interpretable, end-to-end replication of animals–tools EEG decoding that intentionally avoids deep end-to-end models and emphasizes transparent signal processing. Using eventwise segmentation (10,800 trials), EEGLAB preprocessing, two segmentation schemes (BPF and DWT/MODWT), and a compact 5-band  $\times$  15-feature representation, we trained standard classifiers and evaluated on a held-out test set. BPF slightly outperformed DWT (best AUC 0.572 vs. 0.556), and importance analyses localized discriminative information primarily to alpha/delta bands over centro–parietal and fronto–central regions—consistent with known semantic ERP topographies. Absolute accuracy remains modest, underscoring the challenge of single-trial semantic decoding from scalp EEG and pointing to representation and cross-subject variability as key limiting factors.

The resulting pipeline is simple to reproduce, easy to extend, and suitable as a baseline for future work. We recommend complementing it with subject-wise validation, richer spatial/time–frequency representations, and calibrated decision thresholds. Together, these steps should help close the gap between interpretable pipelines and state-of-the-art perfor-

mance while preserving the clarity needed for reliable scientific comparison and eventual BCI translation.

## ACKNOWLEDGMENT

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As noted in prior analyses [1], [2], ... Dataset: OpenNeuro ds004514 [4]. Preprocessing reference: [3]. Related neural decoding work: [5]–[7].

## REPRODUCIBILITY STATEMENT

EEGLAB command history (eeghist), batch scripts, and feature-extraction code (BPF and MODWT-MRA) are provided in the project repository: [your-link](#). All results were produced with CAR, 0.5–45 Hz BPF, CleanLine (50 Hz), extended Infomax ICA, ICLabel (threshold 0.80), and a stratified 80/20 train/test split (seed 42). Feature scaling used training-only  $z$ -scores.